Task 15: Session 15(AIML)-Assignment

Name :- Harsh Nilesh Chaturkar

Domain :- AIML

Collage :- Government Polytechnic Yavatmal

Branch :- Computer Engineering

Github link :- https://github.com/harshchaturkar/Mini-Project-repo.git

---

1. Import the Pandas library

[1] 0s

```
import numpy as np
import pandas as pd
```

[2] 14s

```
from google.colab import files
uploaded = files.upload()
```

```
Choose files Countries.csv
Countries.csv(text/csv) - 95410 bytes, last modified: 04/07/2026 - 100% done
Saving Countries.csv to Countries.csv
```

2. Load the csv dataset into a Dataframe

```
filepath = "/content/Countries.csv"
df = pd.read_csv(filepath)
```

```
df
```

	country	country_long	currency	capital_city	region	continent	demonym	latitude	longitude	agricultural_land	...	population	women_parliament_seats_pct
0	Afghanistan	Islamic State of Afghanistan	Afghan afghani	Kabul	Southern Asia	Asia	Afghan	33.000000	65.000000	383560.0	...	41128771	27.01610
1	Albania	Republic of Albania	Albanian lek	Tirana	Southern Europe	Europe	Albanian	41.000000	20.000000	11655.5	...	2775634	35.71430
2	Algeria	People's Democratic Republic of Algeria	Algerian dinar	Algiers	Northern Africa	Africa	Algerian	28.000000	3.000000	413588.0	...	44903225	8.10811
3	Andorra	Principality of Andorra	Euro	Andorra la Vella	Southern Europe	Europe	Andorran	42.500000	1.500000	187.2	...	79824	46.42860
4	Angola	People's Republic of Angola	Angolan kwanza	Luanda	Middle Africa	Africa	Angolan	-12.500000	18.500000	569525.0	...	35588987	33.63640
...	...	...	...	...	...	...	...	...	...	...	...	...	...
189	Vietnam	Socialist Republic of Vietnam	Vietnamese dong	Hanoi	South-Eastern Asia	Asia	Vietnamese	16.166667	107.833333	123600.0	...	98186856	30.26050

3. df.head()

```
df.head()
```

	country	country_long	currency	capital_city	region	continent	demonym	latitude	longitude	agricultural_land	...	population	women_parliament_seats_pct	rural_po
0	Afghanistan	Islamic State of Afghanistan	Afghan afghani	Kabul	Southern Asia	Asia	Afghan	33.0	65.0	383560.0	...	41128771	27.01610	
1	Albania	Republic of Albania	Albanian lek	Tirana	Southern Europe	Europe	Albanian	41.0	20.0	11655.5	...	2775634	35.71430	
2	Algeria	People's Democratic Republic of Algeria	Algerian dinar	Algiers	Northern Africa	Africa	Algerian	28.0	3.0	413588.0	...	44903225	8.10811	
3	Andorra	Principality of Andorra	Euro	Andorra la Vella	Southern Europe	Europe	Andorran	42.5	1.5	187.2	...	79824	46.42860	
4	Angola	People's Republic of Angola	Angolan kwanza	Luanda	Middle Africa	Africa	Angolan	-12.5	18.5	569525.0	...	35588987	33.63640	

5 rows × 64 columns

4) df.tail()

[6]
✓ 0s

```
df.tail()
```

	country	country_long	currency	capital_city	region	continent	demonym	latitude	longitude	agricultural_land	...	population	women_parliament_seats_pct	ru
189	Vietnam	Socialist Republic of Vietnam	Vietnamese dong	Hanoi	South-Eastern Asia	Asia	Vietnamese	16.166667	107.833333	123600.0	...	98186856	30.2605	
190	West Bank and Gaza	West Bank and Gaza	Israeli new shekel	Ramallah	Western Asia	Asia	Palestinian	31.900000	35.200000	4449.0	...	5043612	NaN	
191	Yemen	Republic of Yemen	Yemeni rial	Sana'a	Western Asia	Asia	Yemeni	15.000000	48.000000	234520.0	...	33696614	0.0000	
192	Zambia	Republic of Zambia	New Zambian kwacha	Lusaka	Eastern Africa	Africa	Zambian	-15.000000	30.000000	238360.0	...	20017675	15.0602	
193	Zimbabwe	Republic of Zimbabwe	U.S. dollar	Harare	Eastern Africa	Africa	Zimbabwean	-20.000000	30.000000	162000.0	...	16320537	30.5660	

5 rows × 64 columns

5) df.shape

[9]
✓ 0s

```
df.shape
```

(194, 64)

6) df.columns

```
[10]
 0s
df.columns

Index(['country', 'country_long', 'currency', 'capital_city', 'region',
       'continent', 'demonym', 'latitude', 'longitude', 'agricultural_land',
       'forest_area', 'land_area', 'rural_land', 'urban_land',
       'central_government_debt_pct_gdp', 'expense_pct_gdp', 'gdp',
       'inflation', 'self_employed_pct', 'tax_revenue_pct_gdp',
       'unemployment_pct', 'vulnerable_employment_pct',
       'electricity_access_pct', 'alternative_nuclear_energy_pct',
       'electricty_production_coal_pct',
       'electricty_production_hydroelectric_pct',
       'electricty_production_gas_pct', 'electricty_production_nuclear_pct',
       'electricty_production_oil_pct', 'electricty_production_renewable_pct',
       'energy_imports_pct', 'fossil_energy_consumption_pct',
       'renewable_energy_consumption_pct', 'co2_emissions',
       'methane_emissions', 'nitrous_oxide_emissions',
       'greenhouse_other_emissions', 'urban_population_under_5m',
       'health_expenditure_pct_gdp', 'health_expenditure_capita',
       'hospital_beds', 'hiv_incidence', 'suicide_rate', 'armed_forces',
       'internally_displaced_persons', 'military_expenditure_pct_gdp',
       'birth_rate', 'death_rate', 'fertility_rate', 'internet_pct',
       'life_expectancy', 'net_migration', 'population_female',
       'population_male', 'population', 'women_parliament_seats_pct',
       'rural_population', 'urban_population', 'press', 'democracy_score',
       'democracy_type', 'median_age', 'political_leader', 'title'],
      dtype='object')
```

7) df.info()

```
[11]
 0s
df.info()

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 194 entries, 0 to 193
Data columns (total 64 columns):
 #   Column                                   Non-Null Count  Dtype
---  ------                                   --------------  -----
 0   country                                  194 non-null    object
 1   country_long                             194 non-null    object
 2   currency                                 194 non-null    object
 3   capital_city                             194 non-null    object
 4   region                                   194 non-null    object
 5   continent                                194 non-null    object
 6   demonym                                  194 non-null    object
 7   latitude                                 194 non-null    float64
 8   longitude                                194 non-null    float64
 9   agricultural_land                        193 non-null    float64
 10  forest_area                              194 non-null    float64
 11  land_area                                194 non-null    float64
 12  rural_land                               194 non-null    float64
 13  urban_land                               194 non-null    float64
 14  central_government_debt_pct_gdp          120 non-null    float64
 15  expense_pct_gdp                          156 non-null    float64
 16  gdp                                      193 non-null    float64
 17  inflation                                184 non-null    float64
 18  self_employed_pct                        179 non-null    float64
 19  tax_revenue_pct_gdp                      159 non-null    float64
 20  unemployment_pct                         179 non-null    float64
 21  vulnerable_employment_pct                179 non-null    float64
 22  electricity_access_pct                   194 non-null    float64
 23  alternative_nuclear_energy_pct           138 non-null    float64
```

8) df.describe()

[12] 0s

```
df.describe()
```

	latitude	longitude	agricultural_land	forest_area	land_area	rural_land	urban_land	central_government_debt_pct_gdp	expense_pct_gdp	gdp	...	n
count	194.000000	194.000000	1.930000e+02	1.940000e+02	1.940000e+02	1.940000e+02	194.000000	120.000000	156.000000	1.930000e+02	...	
mean	18.975601	22.027491	2.454551e+05	2.086784e+05	6.675087e+05	6.563711e+05	9777.116531	66.759366	30.051403	5.144851e+11	...	
std	23.876225	66.396389	6.356268e+05	7.824926e+05	1.837107e+06	1.811169e+06	42301.458421	71.806247	26.740880	2.307148e+12	...	
min	-41.000000	-175.000000	4.000000e+00	0.000000e+00	2.027000e+00	3.495450e-02	0.000000	0.000000	0.000267	6.034940e+07	...	-5
25%	4.000000	-5.000000	6.464000e+03	3.331775e+03	2.355250e+04	2.186562e+04	359.618250	31.951300	18.371875	1.181390e+10	...	-
50%	16.583333	21.500000	3.872780e+04	2.528925e+04	1.203750e+05	1.159945e+05	1645.170000	55.426850	27.337750	4.115390e+10	...	
75%	40.000000	50.162500	2.150000e+05	1.236735e+05	5.237000e+05	4.911505e+05	4054.022500	79.539300	35.083500	2.519450e+11	...	
max	65.000000	178.000000	5.285080e+06	8.153120e+06	1.637690e+07	1.622420e+07	522345.000000	687.994000	310.443000	2.546270e+13	...	5

8 rows × 54 columns

9) df.dtypes

[13] 0s

```
df.dtypes
```

	0
country	object
country_long	object
currency	object
capital_city	object
region	object
...	...
democracy_score	float64
democracy_type	object
median_age	float64
political_leader	object
title	object

64 rows × 1 columns

dtype: object

10) df.isnull().sum()

[14]
0s

```
df.isnull().sum()
```

	0
country	0
country_long	0
currency	0
capital_city	0
region	0
...	...
democracy_score	0
democracy_type	0
median_age	0
political_leader	7
title	7

64 rows × 1 columns

dtype: int64

11) df.nunique()

[15]
0s

```
df.nunique()
```

	0
country	194
country_long	194
currency	142
capital_city	194
region	22
...	...
democracy_score	146
democracy_type	5
median_age	143
political_leader	187
title	17

64 rows × 1 columns

dtype: int64

12) df.sample(5)

```
[16] df.sample(5)
```

	country	country_long	currency	capital_city	region	continent	demonym	latitude	longitude	agricultural_land	...	population	women_parliament_seats_pct	ru
18	Bhutan	Kingdom of Bhutan	Bhutanese ngultrum	Thimphu	Southern Asia	Asia	Bhutanese	27.5	90.5	5130.0	...	782455	17.39130	
141	Rwanda	Republic of Rwanda	Rwandan franc	Kigali	Eastern Africa	Africa	Rwandan	-2.0	30.0	18117.0	...	13776698	61.25000	
177	Turkey	Republic of Turkey	New Turkish lira	Ankara	Western Asia	Asia	Turkish	39.0	35.0	377620.0	...	85341241	17.35400	
20	Bosnia and Herzegovina	Bosnia and Herzegovina	Bosnia and Herzegovina convertible mark	Sarajevo	Southern Europe	Europe	Bosnian	44.0	18.0	22160.0	...	3233526	16.66670	
77	Iran	Islamic Republic of Iran	Iranian rial	Tehran	Southern Asia	Asia	Iranian	32.0	53.0	470130.0	...	88550570	5.59441	

5 rows × 64 columns

13) df.value_counts()

```
[17] df.value_counts()
```

country	country_long	currency	capital_city	region	continent	demonym	latitude	longitude	agricultural_land	forest_area	land_area	rural_land	urban_la
Azerbaijan	Republic of Azerbaijan	New Azeri manat	Baku	Western Asia	Asia	Azerbaijani	40.500000	47.500000	47801.00	11317.7	82646.0	82218.70	2589.8
Bangladesh	People's Republic of Bangladesh	Bangladeshi taka	Dhaka	Southern Asia	Asia	Bangladeshi	24.000000	90.000000	99010.00	18834.0	130170.0	79328.90	56970.0
Bolivia	Plurinational State of Bolivia	Bolivian Boliviano	Sucre	South America	Americas	Bolivian	-17.000000	-65.000000	377870.00	508338.0	1083300.0	1059510.00	1767.5
Brazil	Federative Republic of Brazil	Brazilian real	Brasília	South America	Americas	Brazilian	-10.000000	-55.000000	2368790.00	4966200.0	8358140.0	8385500.00	45853.3
Cameroon	Republic of Cameroon	Central African CFA franc	Yaoundé	Middle Africa	Africa	Cameroonian	6.000000	12.000000	97500.00	203405.0	472710.0	463837.00	2352.11
Colombia	Republic of Colombia	Colombian peso	Bogotá	South America	Americas	Colombian	4.000000	-72.000000	482428.00	591419.0	1109500.0	1127200.00	5114.5
Congo	Republic of Congo	Central African CFA franc	Brazzaville	Middle Africa	Africa	Congolese	-1.000000	15.000000	106280.00	219460.0	341500.0	340341.00	420.26
Croatia	Republic of Croatia	Croatian kuna	Zagreb	Southern Europe	Europe	Croatian	45.166667	15.500000	15050.00	19391.1	55960.0	54777.10	1648.41